Parametric Density Estimation I

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Outline

- Density Estimation
- Maximum Likelihood vs Bayesian Learning
- Maximum Likelihood Estimation
- Maximum A-Posteriori (MAP) Estimation
- Example: The Gaussian Case
- Bayesian Learning
- Recursive Bayesian Incremental Learning

Estimation of PDF

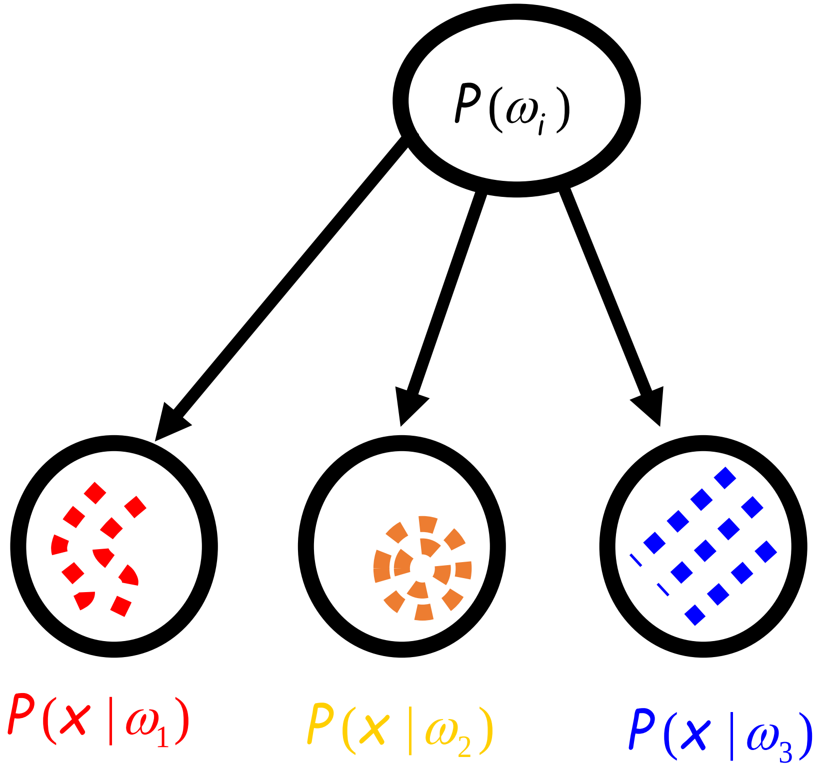
- To get an optimal classifier we need $p(w_i)$ and $p(x|w_i)$, but usually we do not know them.
- Solution: to use training data to estimate the unknown probabilities.
- Key task: Estimation of class conditional density

Learning of PDF

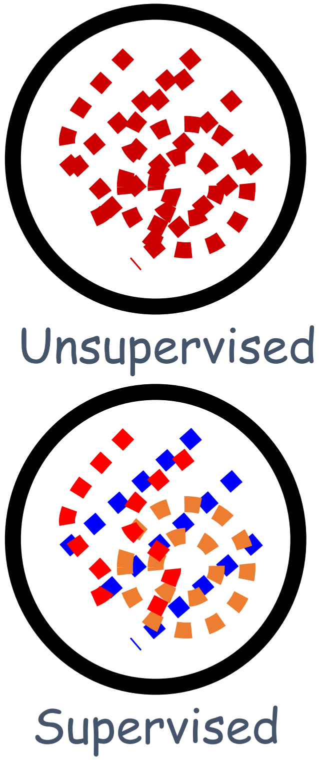
- Supervised learning: we get to see samples from each of the classes "separately" (called tagged or labeled samples).
- Tagged samples are "expensive".
- Two methods: parametric (easier) and non-parametric (harder)

Learning From Observed Data

Hidden



Observed



Parametric Learning

- Assume specific parametric distributions with parameters:
  $p(x|\omega_i) \approx p(x|\theta_i), \theta_i \in \Theta \subset R^p$

- Estimate parameters $\hat{\theta}(D)$ from training data D

- Replace true value of class-conditional density with approximation and apply the Bayesian framework for decision making.

Parametric Learning

- Suppose we can assume that the relevant (class-conditional) densities are of some parametric form. That is,

$$p(x|\omega) \approx p(x|\theta), where\ \theta \in \Theta \in R^p$$

- Examples of parameterized densities:
  - Binomial: x has m 1's and $(n-m)$ 0's

$$p(x|\theta) = \binom{n}{m}\theta^m(1-\theta)^{n-m}, \Theta = [0,1]$$

  - Exponential: Each data point x is distributed according to

$$p(x|\theta) = \theta e^{-\theta x}, \Theta = (0, \infty)$$

  - Normal:

$$p(x|\theta) \sim N(\mu, \sigma^2)$$

Parametric Learning

- Bayesian Decision

$$arg \max_{i} p(\omega_i|x)$$

- Maximum Likelihood Estimation

$$arg \max_{\theta_i} p(D|\theta_i)$$

- Maximum A-Posteriori Estimation

$$arg \max_{\theta_i} p(\theta_i|D), \theta_i = constant$$

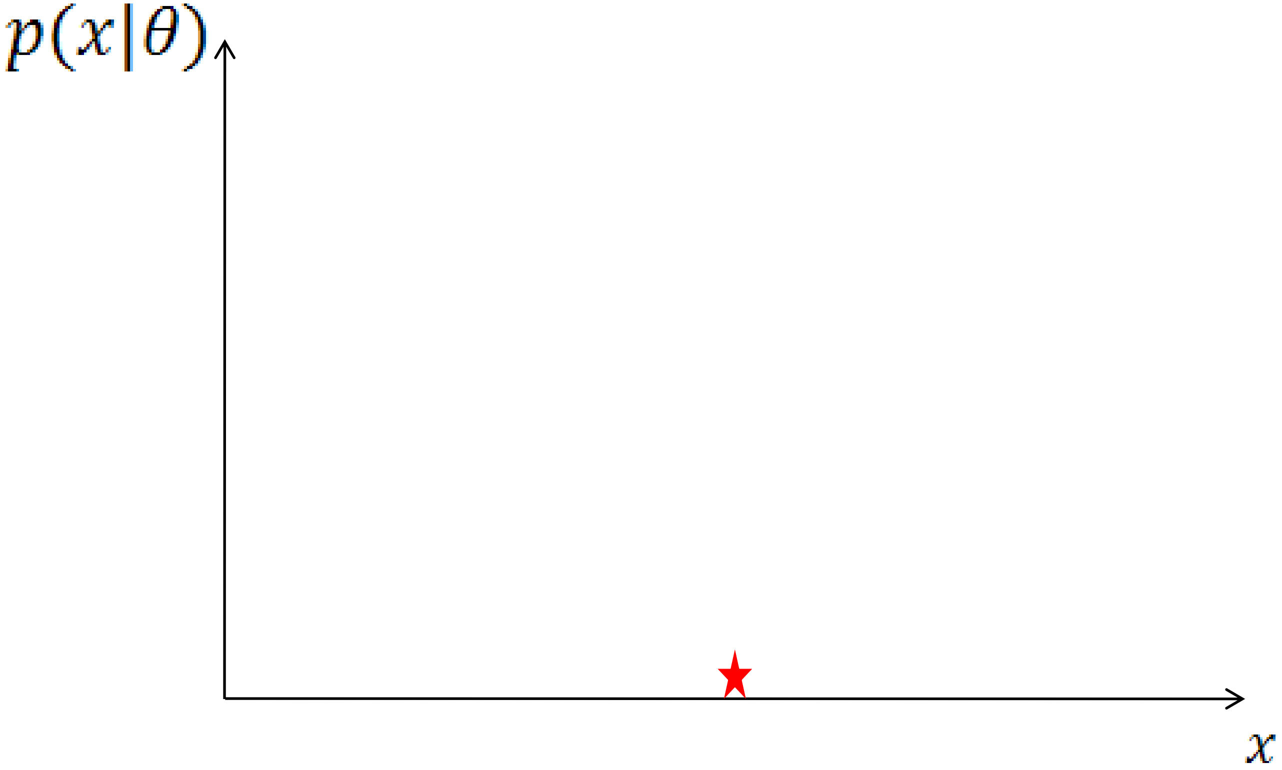
- Bayesian Learning (Estimation)

Find $p(\theta_i|D), \theta_i = random\ variable$

Preliminary Examples

- Maximum Likelihood Estimation

$$\arg\max_{\theta_i} p(D|\theta_i)$$



Preliminary Examples

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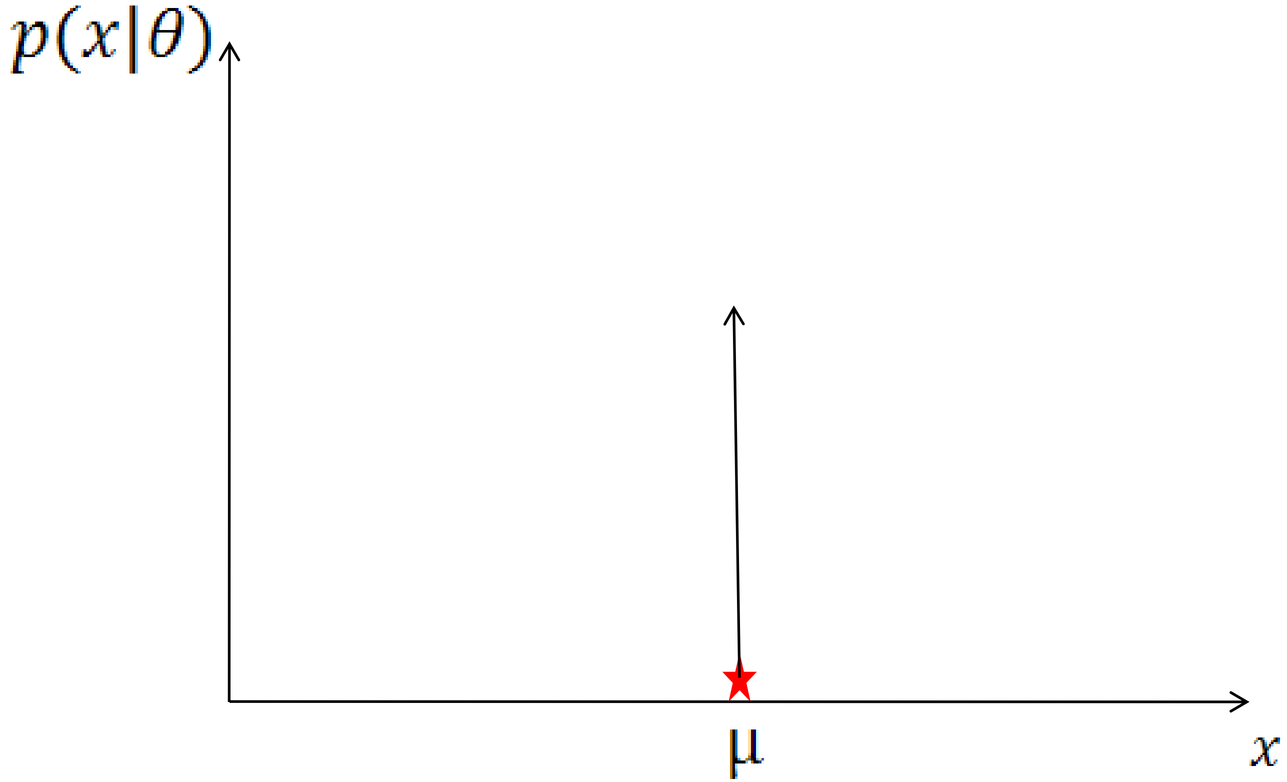

Preliminary Examples

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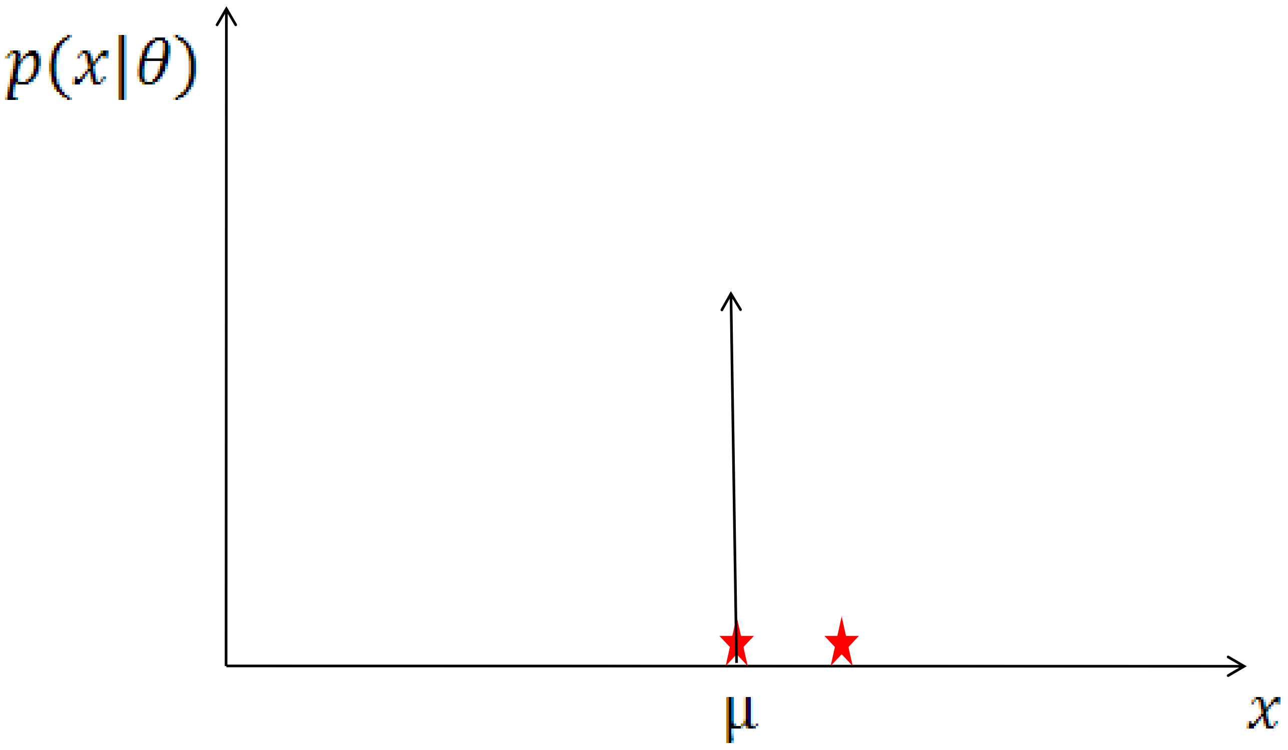

Preliminary Examples

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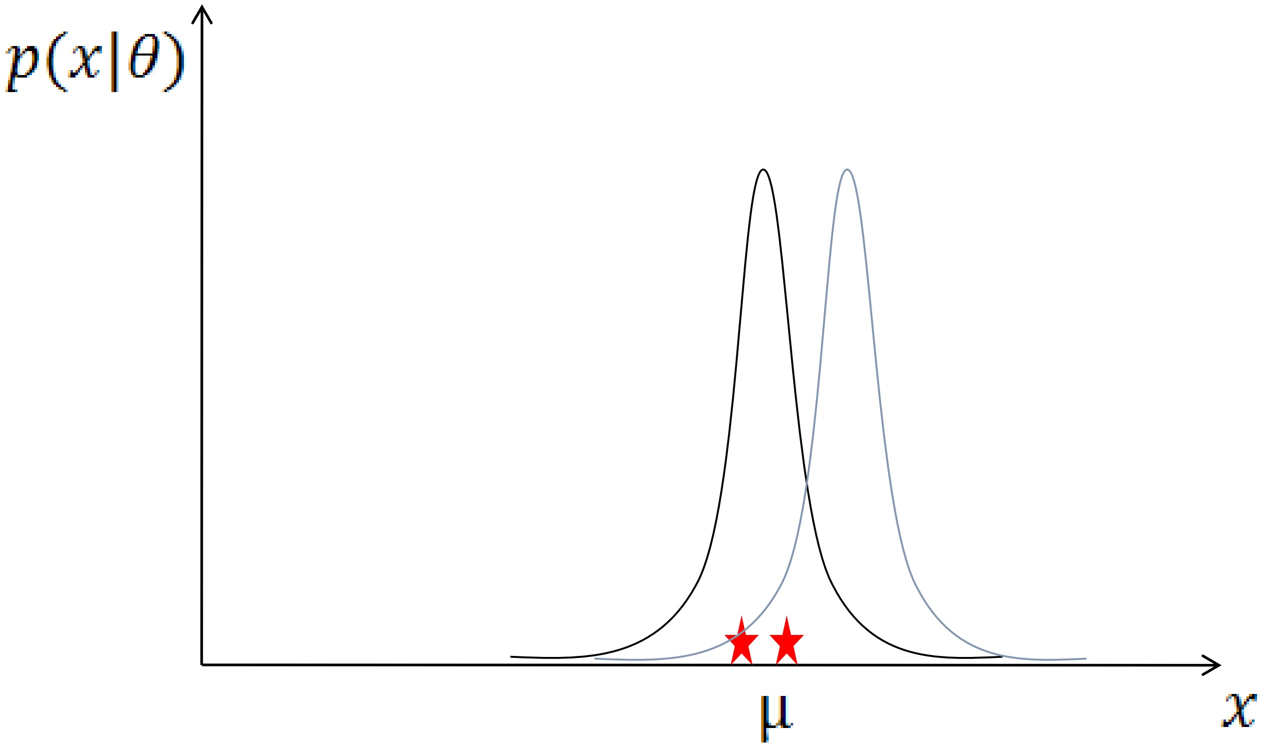

Preliminary Examples

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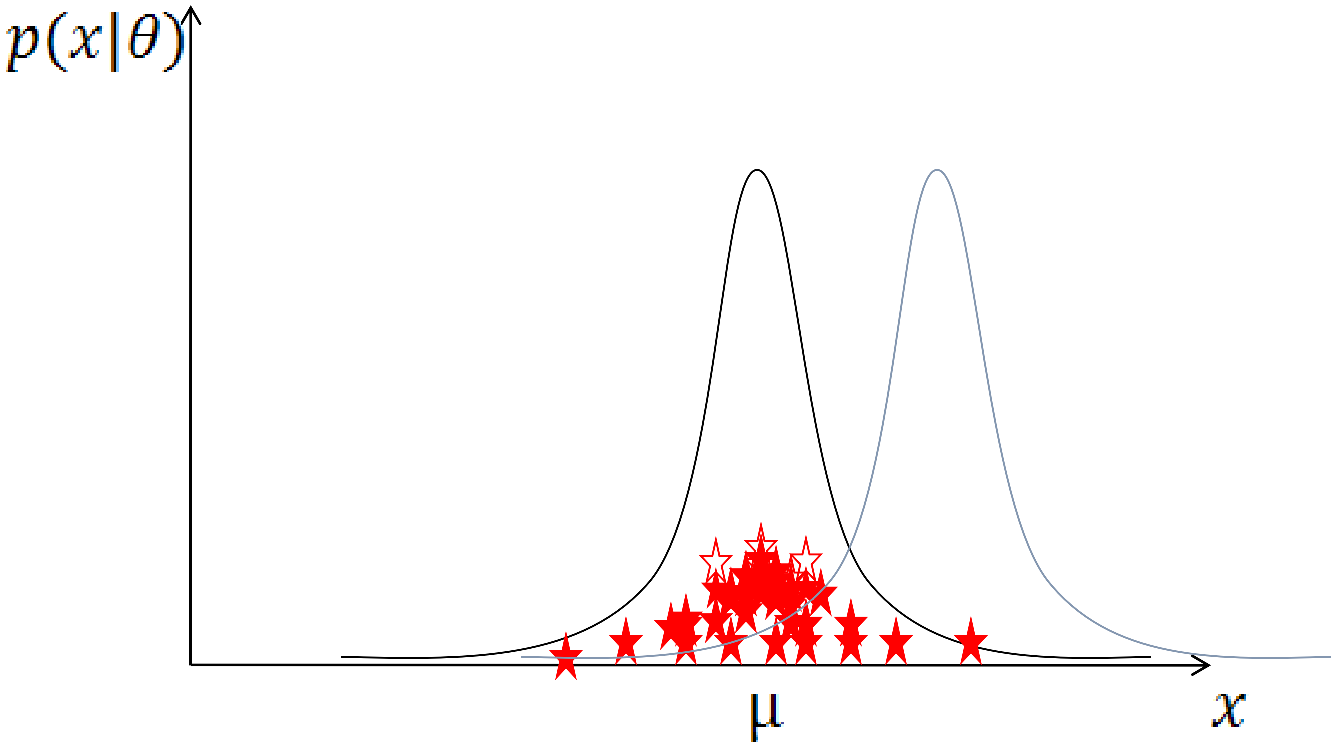
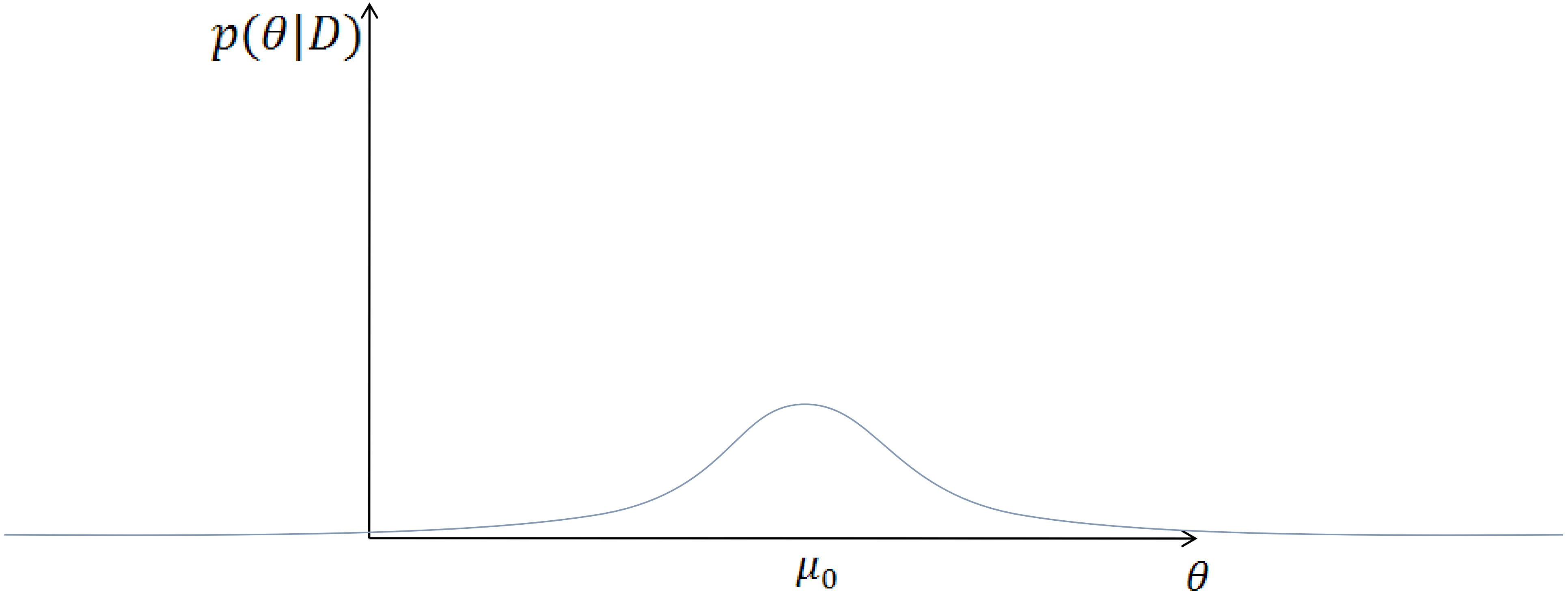


Preliminary Examples

- Bayesian Learning (Estimation)
  Find $p(\theta_i|D), \theta_i = random\ variable$



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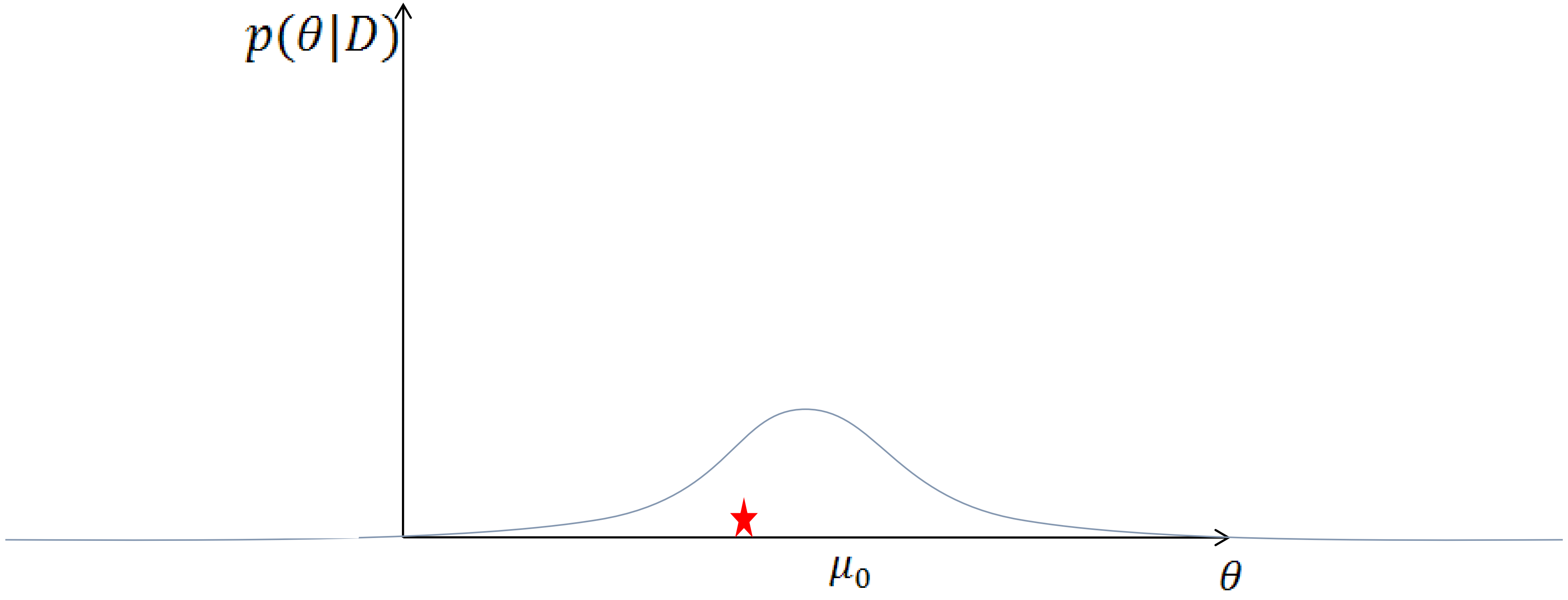

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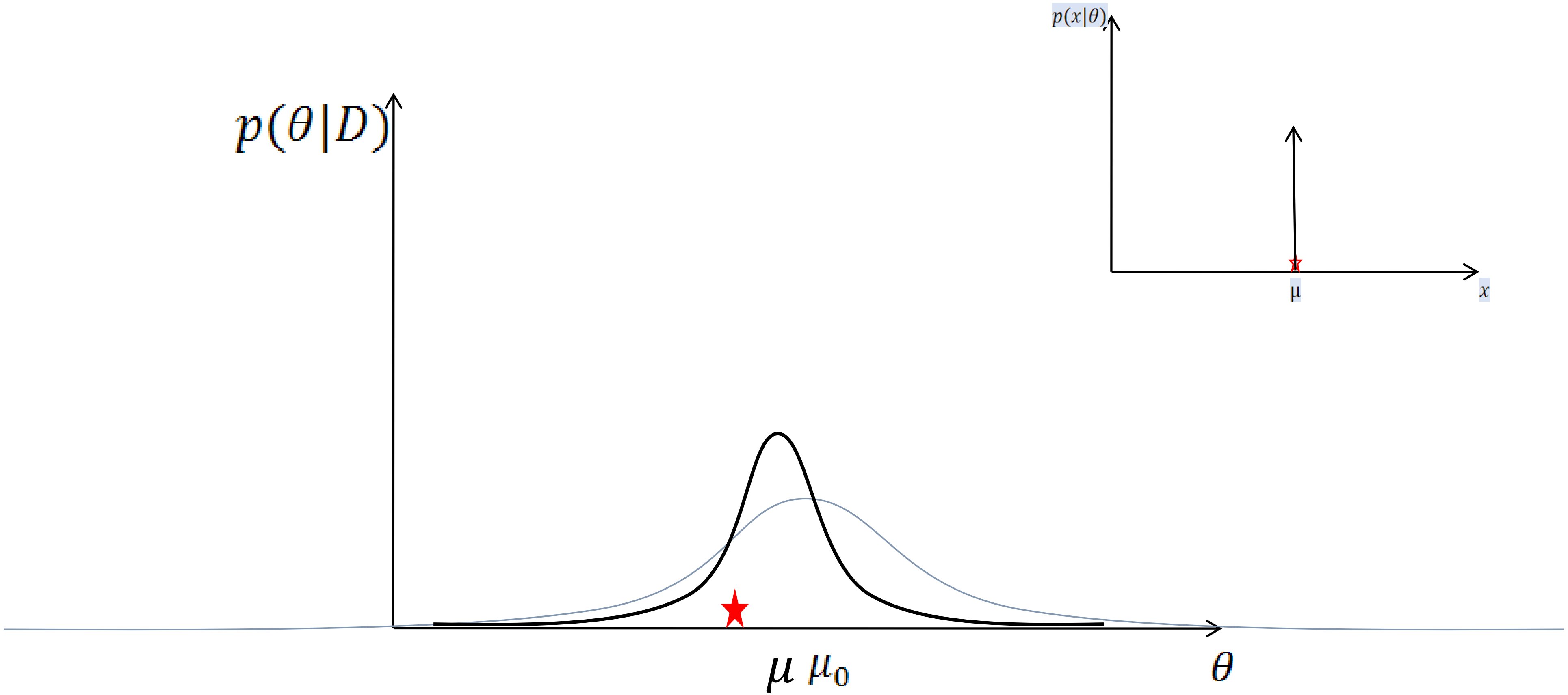

Preliminary Examples

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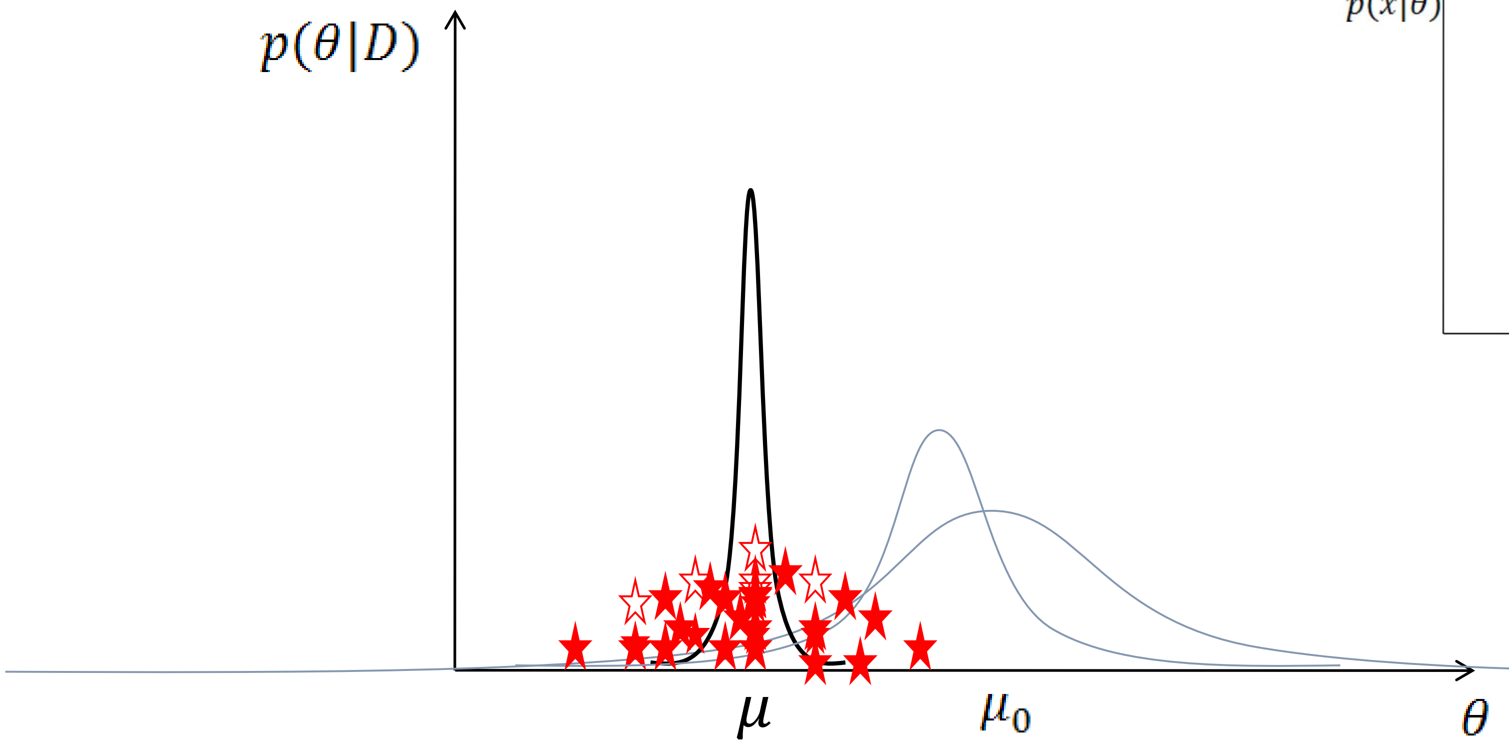
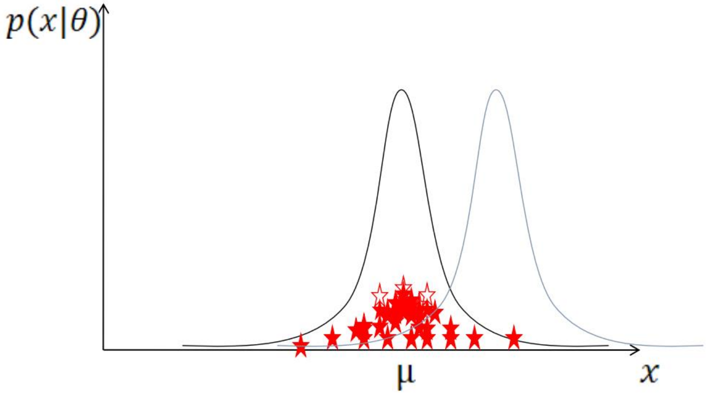




Maximum Likelihood vs Bayesian Learning

- Maximum likelihood estimation: estimate parameter value $\hat{\theta}(D)$ that makes the data most probable (i.e., maximizes the probability of obtaining the sample that has actually been observed),

$$p(x|D) \approx p(x|\hat{\theta}(D)), \hat{\theta}(D) = arg\max_{\theta} p(D|\theta)$$

- Bayesian learning: define a prior probability on the model space $p(\theta)$ and compute the posterior $p(\theta|D)$. Additional samples sharp the posterior density which peaks near the true values of the parameters .

Sampling Model

- It is assumed that a training set $S = \{(x_l, \bar{\omega}_l) | l = 1, \ldots, N\}$ with independently generated samples is available.
- The sample set is partitioned into separate sample sets for each class, $D_j = \left\{x_l \middle| (x_l, \bar{\omega}_l) \in S_j\right\}$
- A generic sample set will simply be denoted by

$$D = \{x_l | l = 1, \ldots, N\}$$

- Each class-conditional $p(x|\omega_j)$ is assumed to have a known parametric form and is uniquely specified by a parameter (vector) θ_j.
- Samples in each set D_j are assumed to be independent and identically distributed (i.i.d.) according to some true probability law $p(x|\omega_j)$.

Log-Likelihood function and Score Function

- The sample sets are assumed to be functionally independent, so the training set S_j contains no information about θ_i for $i \neq j$
- The i.i.d. assumption implies that

$$p\left(D_j \middle| \theta_j\right) = \prod_{x \in D_j} p(x|\theta_j)$$

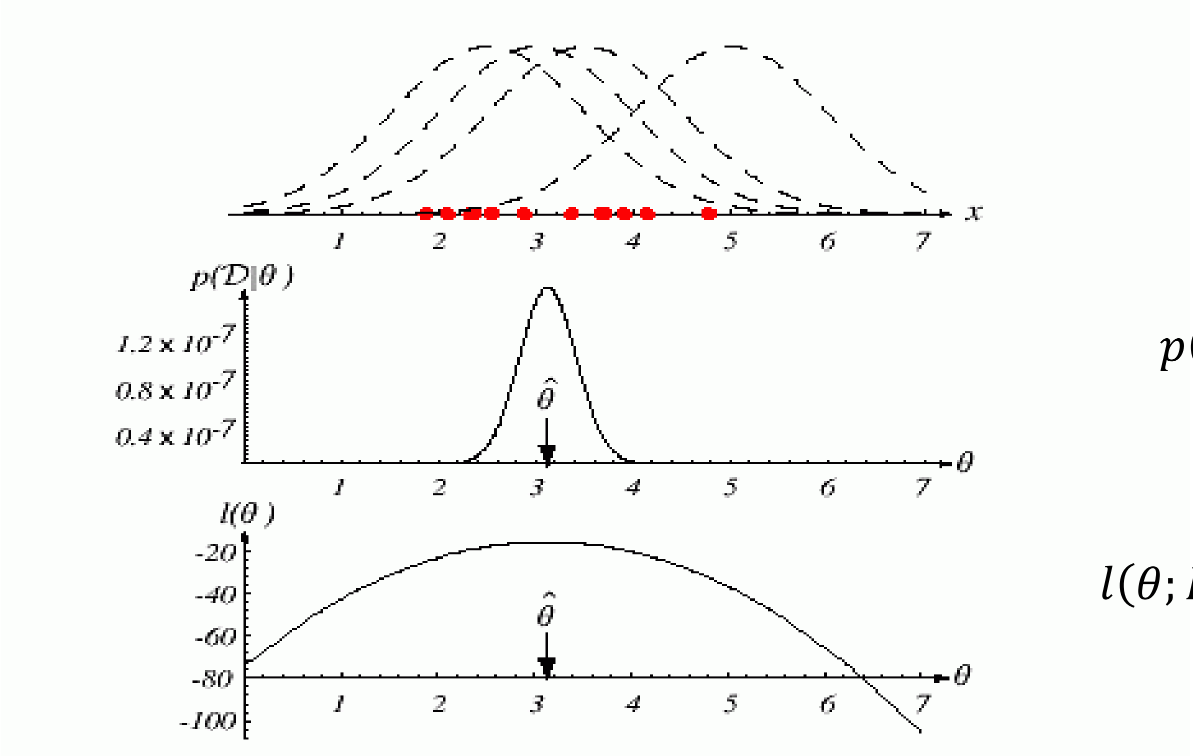
- Let D be a generic sample set of size $n = |D|$
- Log-likelihood function:

$$l(\theta; D) \equiv \ln p(D|\theta) = \sum_{k=1}^{n} \ln p(x_k|\theta)$$

- The log-likelihood function is identical to the logarithm of the probability density (or mass) function, but is interpreted as a function of parameter θ

Log-Likelihood Illustration

- Assume that all the points in D are drawn from some (one-dimensional) normal distribution with some (known) variance and unknown mean.



$$p(D|\theta) = \prod_{x \in D} p(x|\theta)$$

$$l(\theta; D) \equiv \ln p(D|\theta) = \sum_{k=1}^{n} \ln p(x_k|\theta)$$

Maximum Likelihood Estimation (MLE)

- The “most likely value” for MLE is given by

$$\hat{\theta}(D) = \underset{\theta \in \Theta}{argmax}\, l(\theta; D)$$

- Gradient function

$$\nabla_\theta l(\theta; D) = \left[\frac{\partial l(\theta; D)}{\partial \theta_1}, \dots, \frac{\partial l(\theta; D)}{\partial \theta_p}\right]$$

- Necessary condition for MLE (if not on border of domain Θ)

$$\nabla_\theta l(\theta; D) = 0$$

Maximum A-Posteriori (MAP) Estimation

- The "most likely value" for MAP is given by

$$
\begin{aligned}
\hat{\theta}(D) &= arg\max_{\theta \in \Theta} \; p(\theta|D) \\
&= arg\max_{\theta \in \Theta} \; p(D|\theta)\, p_0(\theta)/p(D) \\
&= arg\max_{\theta \in \Theta} \; \ln \prod_k p(x_k|\theta)\, p_0(\theta) \\
&= arg\max_{\theta \in \Theta} \left(\sum_k \ln p(x_k|\theta) + \ln p_0(\theta) \right)
\end{aligned}
$$

- We can ignore the normalizing factor $p(D)$

Example of Maximum Likelihood

The Gaussian Case: Unknown Mean

- Suppose that the samples are drawn from a multivariate normal population with mean μ, and covariance matrix Σ.
- By MLE, find unknown parameter $\theta = \mu$ for sample data

$$D = \{x_k | k = 1, \dots, n\}$$

Example of Maximum Likelihood

The Gaussian Case: Unknown Mean

- Log-likelihood is

  $$l(\mu; D) = \ln p(D|\mu) = \ln \prod_{k=1}^{n} p(x_k|\mu) = \sum_{k=1}^{n} \ln p(x_k|\mu)$$

  where

  $$\ln p(x_k|\mu) = -\frac{1}{2}(x_k - \mu)^t \Sigma^{-1}(x_k - \mu) - \frac{d}{2}\ln 2\pi - \frac{1}{2}\ln|\Sigma|$$

- For a sample point x_k, we have

  $$\nabla_\mu \ln p(x_k|\mu) = \Sigma^{-1}(x_k - \mu)$$

- The maximum likelihood estimate for μ must satisfy

  $$\sum_{k=1}^{n} \Sigma^{-1}(x_k - \mu) = 0 \longrightarrow \mu = \frac{1}{n}\sum_{k=1}^{n} x_k \quad \text{(sample mean)}$$

Example of Maximum Likelihood

The Gaussian Case: Unknown Mean and Covariance

- In the general multivariate normal case, neither the mean nor the covariance matrix is known.
- Estimate the mean and covariance from data.

Example of Maximum Likelihood

The Gaussian Case: Unknown Mean and Covariance

- In the general multivariate normal case, neither the mean nor the covariance matrix is known.
- Consider first the univariate case with $\theta_1 = \mu, \theta_2 = \sigma^2$
- The log-likelihood of a single point is

$$\ln p(x_k|\theta) = -\frac{1}{2}\ln 2\pi\theta_2 - \frac{1}{2\theta_2}(x_k - \theta_1)^2$$

and its derivative becomes

$$\nabla_\theta l(x_k;\theta) = \nabla_\theta \ln p(x_k|\theta) = \begin{bmatrix} \frac{1}{\theta_2}(x_k - \theta_1) \\ -\frac{1}{2\theta_2} + \frac{(x_k - \theta_1)^2}{2\theta_2^2} \end{bmatrix}$$

Example of Maximum Likelihood

The Gaussian Case: Unknown Mean and Covariance

- Setting the gradient to zero, and using all the sample points, we get the following necessary conditions:

$$\sum_k \frac{1}{\theta_2}(x_k - \theta_1) = 0, \quad \sum_k \left(-\frac{1}{2\theta_2} + \frac{(x_k - \theta_1)^2}{2\theta_2^2}\right) = 0$$

- Optimal Solution becomes

$$\hat{\mu} = \theta_1 = \frac{1}{n}\sum_{k=1}^{n} x_k$$

$$\hat{\sigma}^2 = \theta_2 = \frac{1}{n}\sum_{k=1}^{n}(x_k - \hat{\mu})^2$$

Example of Maximum Likelihood

The Gaussian multivariate case

- For the multivariate case, it is easy to show that the MLE estimates are given by

$$\hat{\mu} = \frac{1}{n}\sum_{k=1}^{n} x_k$$

$$\hat{\Sigma} = \frac{1}{n}\sum_{k=1}^{n}(x_k - \hat{\mu})(x_k - \hat{\mu})^t$$

- The MLE for σ^2 is biased (i.e., the expected value over all data sets of size *n* of the sample variance is not equal to the true variance, that is,

$$E\left[\frac{1}{n}\sum_{k=1}^{n}(x_k - \hat{\mu})^2\right] = \frac{n-1}{n}\sigma^2 \neq \sigma^2$$

https://en.wikipedia.org/wiki/Bias_of_an_estimator

https://en.wikipedia.org/wiki/Variance#Sum_of_uncorrelated_variables_(Bienaym%C3%A9_formula)

Interim Summary and Next Outline

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- Maximum A-Posteriori (MAP) Estimation
- Example of ML: The Gaussian Case
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